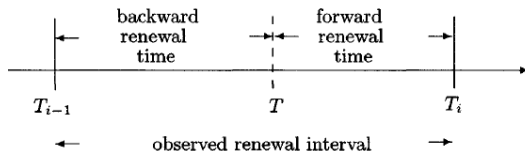


$M/G/1$ queue

Tejas Bodas

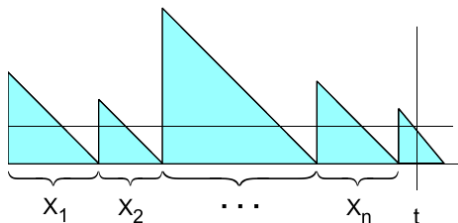
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Age and Residual life of a Renewal process



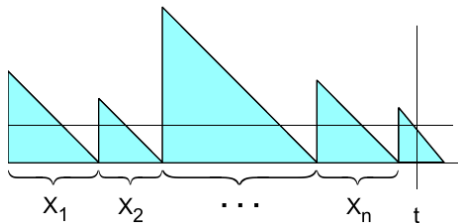
- ▶ Let $A(t)$ and $R(t)$ denote the age and the residual life of the renewal process at time t .
- ▶ Assume you arrive at a Metro station at time t .
- ▶ $A(t)$ is the time since the last metro departed.
- ▶ $R(t)$ is the time till the next Metro arrives.
- ▶ Assume that you arrive uniformly at random to the Metro.
- ▶ What is your average waiting time $\bar{R}$? $\bar{R} = E[X]/2$?

Hitchhiker's Paradox!



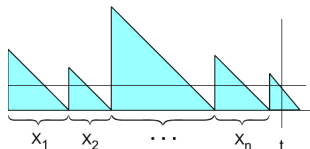
- ▶ Consider $\bar{R} = \lim_{t \rightarrow \infty} \frac{Y(t)}{t}$ where $Y(t) = \int_0^t R(t)$.
- ▶ Using Renewal reward theorem, $\bar{R} = \frac{E[Y]}{E(X)} = \frac{E[X^2]}{2E[X]} \neq E[X]/2$.
- ▶ $\frac{E[X^2]}{2E[X]} = E[X]/2$ only when interarrival times are deterministic.
- ▶ Consider $\bar{A} = \lim_{t \rightarrow \infty} \frac{Y(t)}{t}$ where $Y(t) = \int_0^t A(t)$. $\bar{A}$?
- ▶ What is $\bar{R}$ or $\bar{A}$ when $X_i \sim \exp(\lambda)$?

PASTA



- ▶ The key assumption to Hitchhikers paradox was that you arrive uniformly at random at the busy/metro stop.
- ▶ Now suppose there is a signboard at the metro that tells you the residual time till the next metro.
- ▶ Suppose that you note the residual time after every 5 min interval and compute an empirical average of the residual times.
- ▶ Will this be equal to $\bar{R}$? No!
- ▶ You do not sample $(0, t)$ uniformly.

PASTA



- ▶ What is you make the residual time readings after a random time which is $\exp(\lambda)$ distributed.
- ▶ Your observation process is a $\text{Poisson}(\lambda)$ process.
- ▶ In this case, the empirical average will equal $\bar{R}$.

For a Poisson process, given $N(t) = n$, the arrival times $S_1, \dots, S_n$ have the same distribution as the order statistics of n i.i.d uniform points over $(0, t)$. (Thm 2.3.2 Sheldon ross)

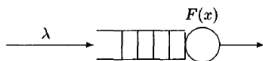
Poisson arrivals see time average! (PASTA)

$M/G/1$ queue



- ▶ Arrival process – Poisson (λ)
- ▶ Single server with unit service rate
- ▶ Arriving jobs require a random service time with arbitrary distribution $F(\cdot)$ with mean b .
- ▶ For an $M/M/1$ queue, $F(\cdot) \sim \exp(\mu)$ and $b = \frac{1}{\mu}$
- ▶ For an $M/G/1$ queue, is $N(t)$ (number of jobs in the system) a Markov chain ? no!

M/G/1 queue



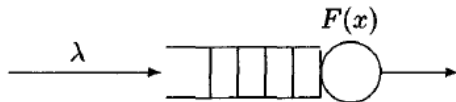
- ▶ General service times do not have the memoryless property.
- ▶ Let $h(x)$ denote the probability that the job will finish service now given that it has received x units of service already. (a.k.a hazard rate)
- ▶ $h(x) = \frac{f(x)}{\bar{F}(x)}$. $h(x) = \mu$ for exponential distribution.
- ▶ Therefore the instantaneous rate of going from $N(t) = n$ to $N(t^+) = n - 1$ depends on the age of the job which is in service at time t .
- ▶ $N(t)$ is therefore not sufficient to describe the evolution of the state.
- ▶ $(N(t), \text{age}(t))$ is a valid descriptor for a Markov chain. We do not study this in the course!

$M/G/1$ queue: Notations



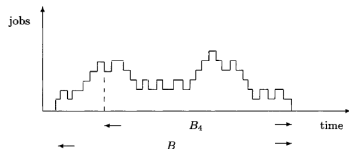
- ▶ L denotes the mean number of jobs in the system.
- ▶ let I denote the mean number of waiting jobs in the queue.
- ▶ $r = L - I$ denotes the mean number of jobs receiving service. This is same as the probability that the server is busy.
- ▶ w denotes the mean time spent by any job waiting for service while W denotes the mean sojourn time.
- ▶ $W = w + b$.

$M/G/1$ queue: Little's law



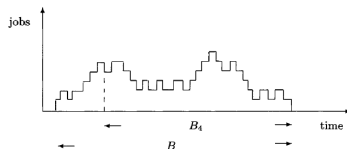
- ▶ $L = I + r$ & $W = w + b$.
- ▶ Using Little's law we have, $L = \lambda W$.
- ▶ Similarly, $I = \lambda w$.
- ▶ This gives us $r = \lambda b$.
- ▶ The probability that the server is busy is λb .
- ▶ Recall that for an $M/M/1$ queue, $1 - \pi(0) = \frac{\lambda}{\mu} = \lambda b$.

Busy period analysis for $M/G/1$



- ▶ What is the mean length of busy period, i.e., $E[B]$?
- ▶ What is the probability that the server is busy? ($1 - \pi_0 = \frac{\lambda}{\mu}$)
- ▶ The time average that the server is busy is $\lim_{t \rightarrow \infty} \frac{1}{t} \int_0^t 1_{\{N(t) > 0\}} dt$.
- ▶ Let $Y(t) = \int_0^t 1_{\{N(t) > 0\}} dt$. This denotes the time for which the server is busy till time t .
- ▶ Using RR theorem, $Y(t)/t$ approaches $\frac{E[B]}{E[B] + \frac{1}{\lambda}}$.

Busy period analysis for $M/G/1$



- ▶ Using RR theorem, $Y(t)/t$ approaches $\frac{E[B]}{E[B] + \frac{1}{\lambda}}$
- ▶ Equating the two averages give us $E[B] = \frac{b}{1-\rho}$ where $\rho = \lambda b$.
- ▶ Mean number of jobs served in a busy period
 $n_B = E[B]/b = \frac{1}{1-\rho}$.

Mean value formulas for $M/G/1$

- ▶ $W = w + b$. $L = I + \rho$.
- ▶ Poisson arrival see time average. So consider a tagged arrival.
- ▶ For this job $w = w_0 + A$ where w_0 is the residual service time of job in server and A is the delay due to waiting jobs.
- ▶ $A = Ib = \lambda wb$.
- ▶ What is w_0 ? Use Hitchhikers paradox!
- ▶ $w_0 = \rho \frac{M_2}{2b}$ where M_2 is the second moment of distribution F .
- ▶ $w = \frac{\lambda M_2}{2} + \lambda wb$.
- ▶ $w = \frac{\lambda M_2}{2(1-\rho)}$.
- ▶ These formulas are called Pollaczek-Khinchin formulas.
- ▶ We used Little's law to obtain mean performance metrics!!

For those interested in Honors/DD

1. Stochastic Optimization

- ▶ Bayesian Optimization (Gaussian processes for ML)
- ▶ Reinforcement learning (Markov Decision Process under uncertainty)
- ▶ Multi-arm bandit optimization (UCB, Thompson, Gittins index)
- ▶ Probabilistic Machine learning

2. Operations Research

- ▶ Performance modeling (this course)
- ▶ Pricing (Data driven approaches, estimating distributions)
- ▶ Choice modeling, Assortment Optimization, Recommender Systems

3. Financial Engineering

- ▶ Portfolio Optimization
- ▶ Optimal stopping for Option pricing
- ▶ Brownian motion, Black Sholes formula, Stochastic Differential Equations